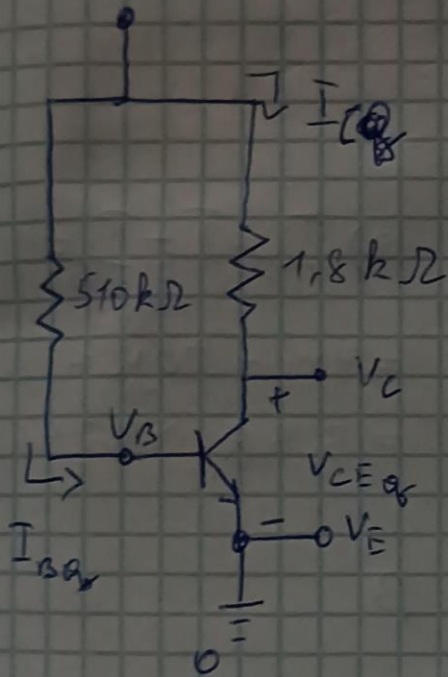


HW N 8

p N 244, EXM



$$a) I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} =$$

$$= \frac{16 - 0,7}{510k} = 30 \mu A$$

$$b) I_{CQ} = \beta I_{BQ} =$$

$$= 120 \cdot 30 = 3,6 mA$$

$$c) V_{CEQ} = V_{CC} - I_{CQ} R_C =$$

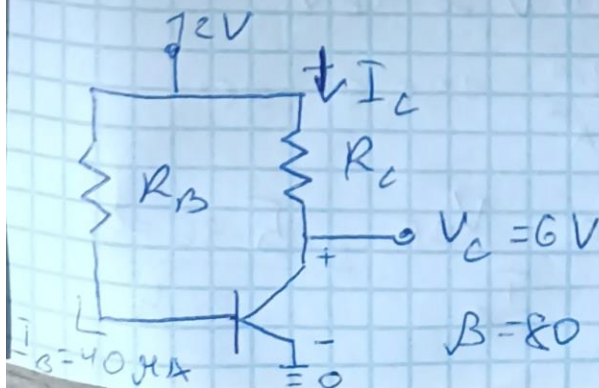
$$= 16 - 3,6 \cdot 1,8 = 9,52V$$

$$d) V_C = V_{CEQ} = 9,52V$$

$$e) V_B = V_{BE} = 0,7V$$

$$f) V_E = 0V$$

p N 244, EX N 2



$$a) I_C = \beta I_B = 80 \cdot 40 =$$

$$= 3,2 mA$$

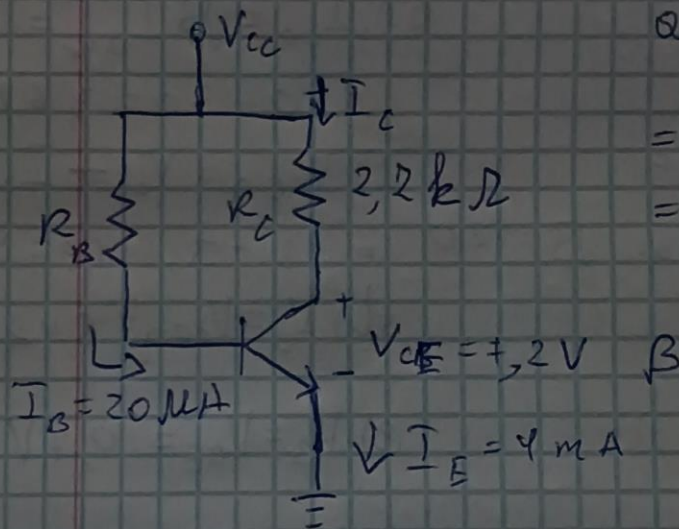
$$b) R_C = \frac{V_{RC}}{I_C} = \frac{V_{CC} - V_C}{I_C} = \frac{12 - 6}{3,2mA} =$$

$$= 1,875 k\Omega$$

$$c) R_B = \frac{V_{RB}}{I_B} = \frac{12 - 0,7V}{40\mu A} = 28,25 \text{ k}\Omega$$

$$d) V_{CE} = V_C = 6V$$

PN 244 ENC N 3



$$a) I_C = I_E - I_B = 4 \text{ mA} - 20 \mu A = 3,98 \text{ mA} \approx 4 \text{ mA}$$

$$b) V_{CC} = V_{CE} + I_C R_C = 7,2 V + 3,98 \text{ mA} \cdot 2,2 \text{ k}\Omega = 15,96 V \approx 16 V$$

$$c) \beta = \frac{I_C}{I_B} = \frac{3,98 \text{ mA}}{20 \mu A} = 199 \approx 200$$

$$d) R_B = \frac{V_{RB}}{I_B} = \frac{V_{CC} - V_{BE}}{I_B} = \frac{15,96 - 0,7}{20 \mu A} = 710 \text{ }\Omega$$

$$= 763 \text{ k}\Omega$$

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$$\bar{I}_{\text{sat}} = \frac{V_{\text{CC}}}{R_{\text{C}}} = \frac{16 \text{ V}}{2,7 \text{ k}\Omega} = 5,93 \text{ mA}$$